



BLOODBIDGE

BRIDGING DONORS. SAVING LIVES.



FIND DONORS

Locate nearby
blood donors easily.



BECOME A DONOR

Register as a donor
and help save lives.



EMERGENCY BLOOD REQUEST

Request blood in
emergencies instantly.



SAFE AND SECURE

Your data is protected
and always secure.

PROJECT TYPE



APP DEVELOPMENT PROJECT



KOTLIN



XML



FIREBASE



ANDROID STUDIO



ANDROID

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LIST OF ACRONYMS

AI	Artificial Intelligence
SOS	Save Our Souls / Emergency Distress Signal
UI	User Interface
UX	User Experience
IDE	Integrated Development Environment
GPS	Global Positioning System
SDK	Software Development Kit
API	Application Programming Interface
XML	Extensible Markup Language
DB	Database
OS	Operating System
SSD	Solid State Drive
CRUD	Create, Read, Update, Delete
OTP	One-Time Password

CHAPTER 1

INTRODUCTION

1.1 Overview

Blood plays a critical role in medical treatments, surgeries, trauma care, and emergency situations. However, hospitals and blood banks often struggle with shortages due to unpredictable demand and slow traditional donor-mobilization methods. Manual systems—such as phone calls, physical records, and local donation drives—are inefficient and may lead to dangerous delays during emergencies.

With the growth of mobile technology, healthcare services are shifting toward faster, digitally enhanced solutions. Mobile applications now offer an effective way to instantly connect blood donors with recipients, ensuring timely availability of blood when it is needed most.

1.2 Specific Details

Our blood donation application- **BloodBridge**, developed using **Kotlin** and **Firebase**, aims to provide a modern, reliable, and user-friendly platform to streamline the donation process. The app allows donors to register easily, store their details securely, and receive real-time notifications for donation requests. Using Firebase, the system ensures fast data synchronization, authentication, and smooth backend operations.

The app enables hospitals and patients to quickly search for eligible donors based on blood group and location, improving response time in emergencies. Information is updated in real-time, allowing donors and recipients to communicate instantly. By integrating modern mobile features, the application makes blood donation more accessible, efficient, and organized.

1.3 Problem statement

Blood donation is often hindered by a lack of timely communication between donors and recipients. Many people willing to donate are unaware of where and when their donation is needed, while hospitals and patients struggle to find donors during emergencies. There is a need for an efficient system that connects donors, recipients, and hospitals quickly and reliably.

Some existing problems:

- User medical details (age, tests like BP, temperature, etc.) taken during the time of blood donation is not integrated into apps for hospital use.

- Existing blood donation apps lack advanced support features like AI assistance.
- There is no option where blood donor can view records/details of their donation.
- Many apps suffer from poor design, crashes, and limited functionality.

1.4 Objectives

- 1) Create a separate interface for public users & hospital users.
- 2) Develop a comprehensive public user interface with options of:
 - Enabling public users to authenticate via email, google or phone
 - Donate, request blood and other features like- times of donations, next donation date, their donation records and history.
 - Develop an AI-powered chatbot to guide donors and recipients with instant responses
- 3) Develop a comprehensive hospital user interface with options of, that:
 - Enables hospital users to authenticate using hospital email address by sending a verification link.
 - Enables hospitals to post an Emergency SOS blood request to donors in critical situations.
 - Enable hospitals to enter and store donor details (age, test results, forms) during the time of blood donation directly into the app.
 - Make the report available to donor on public user side.
- 4) And overall:
 - Provide a clean, simple, and flexible UI for better usability.
 - Fix app crashes and improve stability for a smooth experience.

CHAPTER 2

LITERATURE SURVEY

SL No	Source	Paper Details	Key Attributes/Findings	Challenges/ Limitations
1	IEEE(2024) (ITIKD)	Enhancing Emergency Blood Supply Chain Management Through a Demand-Driven Mobile Application for Rapid Donor Mobilization	The study developed the Prana mobile app to rapidly connect donors and recipients ,improving emergency blood delivery speed. It ensures secure, verified, and donor communication through OTP authentication and Firebase integration.	Manual Verification Required: Hospitals must verify OP numbers and registrations, which adds a human dependency that could slow down automation.
2	IEEE(2024) (INCIP)	A Digital Solution for Coordinating Blood Donations and Transfusions	The study developed a web-based blood transfusion system using ReactJS and Firebase that connects donors, hospitals, and blood banks on a single platform. The system enables real-time coordination, efficient blood request management, and secure authentication.	The model lacks integration with existing hospital or national blood bank systems , limiting interoperability
3	IEEE(2024) (ITIKD)	Blood Bank Management System: Enhancing Security and Transparency in Blood Donation	The study introduces a blockchain-based blood bank management system integrating PHP, MySQL, and Ethereum/IPFS to ensure secure, transparent blood donation processes	Blockchain integration increases computational cost and may face latency issues
4	IEEE(2024) (ICICT)	Blood Delivery Management System using Google Map – Blood Hub	The paper introduces Blood Hub , a real-time web and mobile application designed to connect blood donors, recipients, hospitals, and organizations . Developed with ReactJS, NodeJS, MongoDB, ExpressJS, and Flutter	Relies heavily on, APIs, and external services (like Amazon Connect)

5	IEEE(2024) (EmergIN)	Digital Solutions for Community Health: Enhancing Blood Donation with Mobile Technology	The paper proposes a Flutter-based blood donation app that uses Google Maps GPS to instantly locate nearby donors during emergencies and display their profiles for quick contact. It ensures high data security using Firestore, Cloud Functions, encryption, OTP verification (Twilio)	The system depends heavily on internet , which limits performance in remote areas
6	IEEE(2024) (ICETCI)	RAKTAMITRA- An Alert Centric Blood Donation Initiative	“RAKTAMITRA” is a secure NSS-only platform where only verified students and faculty can register using their official email. The app helps connect donors and recipients quickly using automated alerts, donor lists, and blood request details.	Because the app allows only specific institutional emails for registration, it can be used by only a limited group of people and cannot be used by everyone.
7	IEEE(2024) (OTCON)	RaktFlow - Blood Bank Management and Donation System	E-RaktFlow provides a single platform for blood, oxygen, and ambulance availability, helping users quickly find nearby hospitals, donors, and oxygen stations. The system uses React Native, Django, REST APIs, and PostgreSQL	Donation history depends on the physical donor ID card—if lost, users cannot track their last donation date .
8	IEEE(2023) (INDIACom)	Smart Application for Blood Donation Management in Health Domain	The study proposes an integrated web and mobile system that connects a central blood bank with multiple health centers, enabling real-time tracking of blood availability	The system depends on manual verification at health centers , which may delay account activation and donor participation.

Table 2.1

CHAPTER 3

SYSTEM REQUIREMENTS SPECIFICATION

3.1 Software Requirements

For Developer System:

OS	Windows 10/11, Linux (Ubuntu), or macOS
IDE	Android Studio (latest stable version)
Programming Language	Kotlin
Database & Backend Services	• Firebase (Realtime Database / Firestore) • Firebase Authentication • Firebase Storage (for profile images, medical certificates, etc.)
SDKs & Libraries	• Android SDK • Google Play Services • Google Maps API • Firebase SDKs • Retrofit/Volley (if using external APIs) • Material Design Components • Glide/Picasso (for image loading)
Build Tools	Gradle (latest version supported by Android Studio)
Testing Tools	Android Emulator (for device testing)

Table 3.1

3.2 Hardware Requirements

For Developer System:

Processor/CPU	Intel Core i5/i7 or AMD Ryzen 5/7 (multi-core recommended)
RAM	Minimum 8 GB (16 GB recommended for smooth Android Studio + Emulator use)
Storage	256 GB SSD or higher (SSD strongly recommended for faster build times)

Graphics	Integrated GPU sufficient Dedicated GPU recommended for better Android Emulator performance
Internet Connection	Stable broadband (required for Firebase sync, Google Maps API, dependency downloads)
USB Connectivity	USB 2.0/3.0 ports for testing on physical Android devices

Table 3.2

For End Users:

OS	Android 8.0 (Oreo) or above
RAM	Minimum 2 GB (4 GB or above recommended for lag-free performance)
Storage	At least 100 MB free space for app installation + cached data
CPU/Processor	Any modern dual-core or quad-core processor (ARM-based recommended)
Connectivity:	Internet connection (WiFi or Mobile Data) required for Firebase & Maps GPS enabled for location-based donor search
Sensors:	Location/GPS sensor Network support (4G/5G preferred)

Table 3.3

CHAPTER 4

DESIGN AND METHODOLOGY

4.1 Architecture Diagram

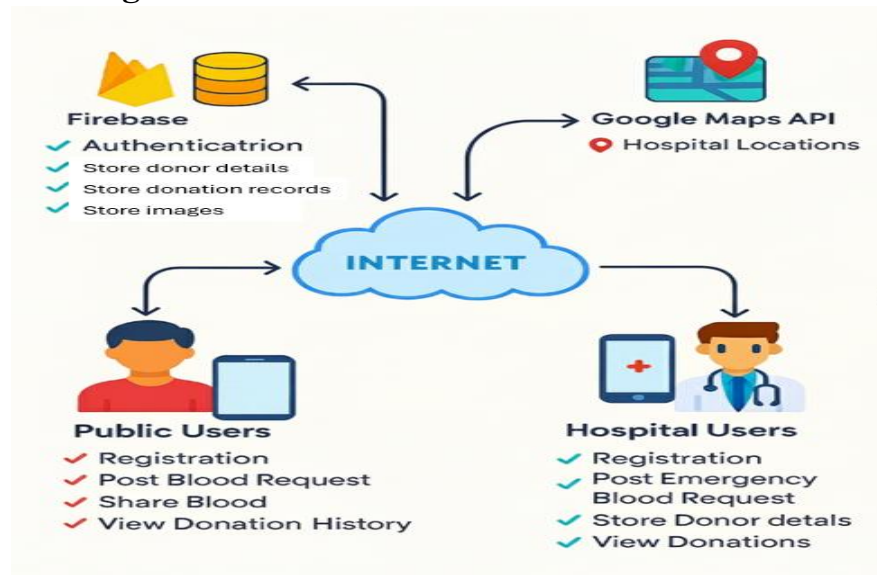


Fig. 4.1

4.2 Use Case Diagram

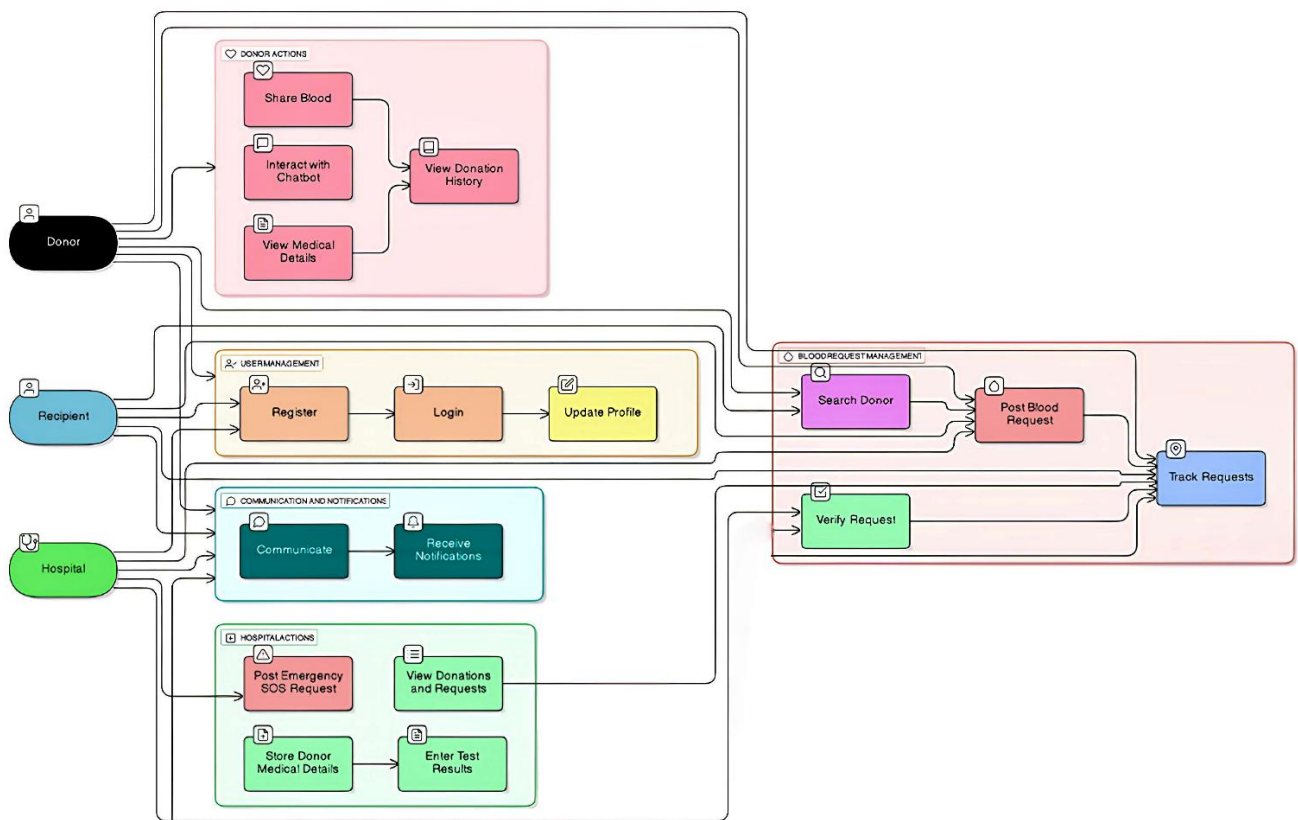


Fig. 4.2

4.3 Dataflow Diagram

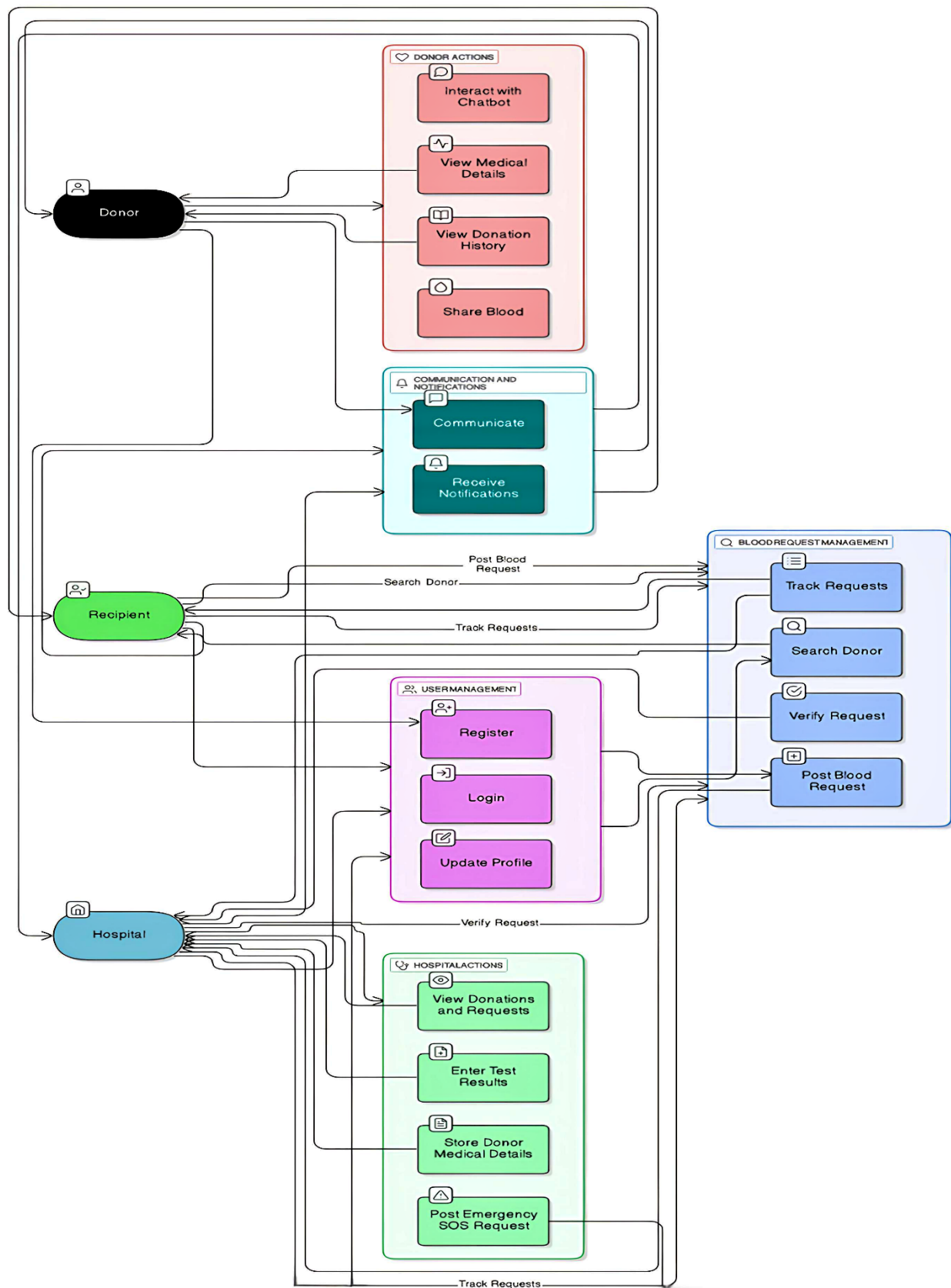


Fig. 4.3

4.4 Methodology

The development of the Blood Donation App followed the **Agile Software Development Methodology**, which emphasizes flexibility, incremental progress, and continuous improvement. Instead of building the entire application at once, the project was divided into **small, manageable development cycles**. Each cycle focused on a specific set of features such as authentication, database connectivity, user interface, or notifications.

At the end of every cycle, the module was tested thoroughly to ensure functionality, stability, and smooth user experience. Feedback from testing was immediately used to refine or improve the module before moving forward. This iterative process allowed issues to be identified early, reduced the chances of large-scale errors, and ensured that the application evolved step by step into a stable and reliable system.

By continuously reviewing progress and integrating completed modules one at a time, the Agile methodology helped maintain clarity, improve development speed, and deliver a well-structured, high-quality blood donation application

CHAPTER 5

IMPLEMENTATION

5.1 Platform Selection

The Blood Donation App was implemented using Android Studio. Android Studio is the official Integrated Development Environment (IDE) for developing Android apps.

- Android Studio is developed by Google and based on IntelliJ IDEA.
- Supports programming in Kotlin, Java, and XML.
- Provides tools for designing app layouts, writing code, debugging, and testing.
- Includes an Android Emulator to run and test apps without a physical phone.

Frontend:

- Developed using Kotlin and XML(eXtensible Markup Language) layouts.
- All screens such as Login, Registration, Dashboard, Donation History, and Request Page are designed using XML layouts.
- Kotlin handles button clicks, data validation, and navigation between screens.
- It also connects with Firebase through the Firebase SDK, sending and receiving data in real time.

Backend:

- Firebase Authentication handles user sign-up and login using email, Google, or phone.
- Firebase Realtime Database stores user details, donation records, and blood requests.

Google Maps Integration

Enabling Google Maps API:

- Google Maps SDK for Android is enabled through the Google Cloud Console.
- An API Key is generated, which allows the app to access Google Maps services.

Sample Code Implementation Snippets

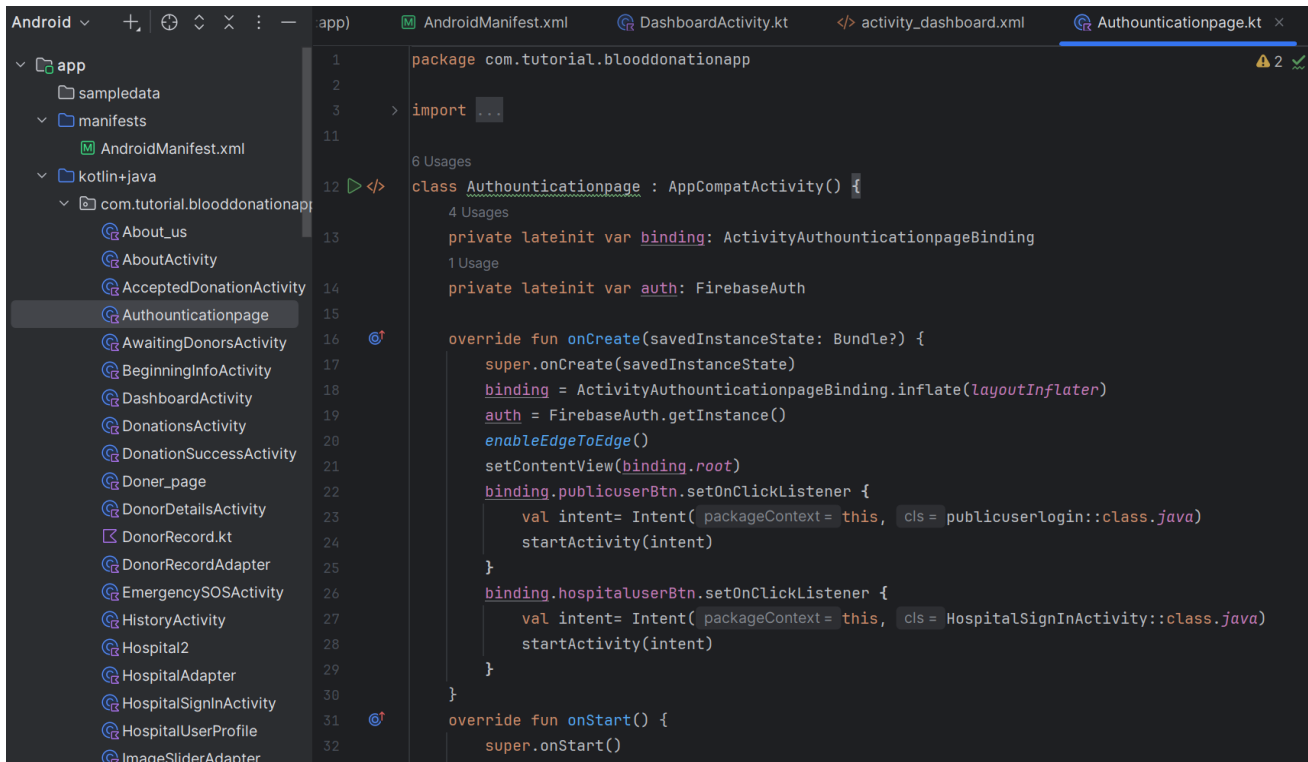


Fig. 5.1 – Kotlin logic code of Login options page

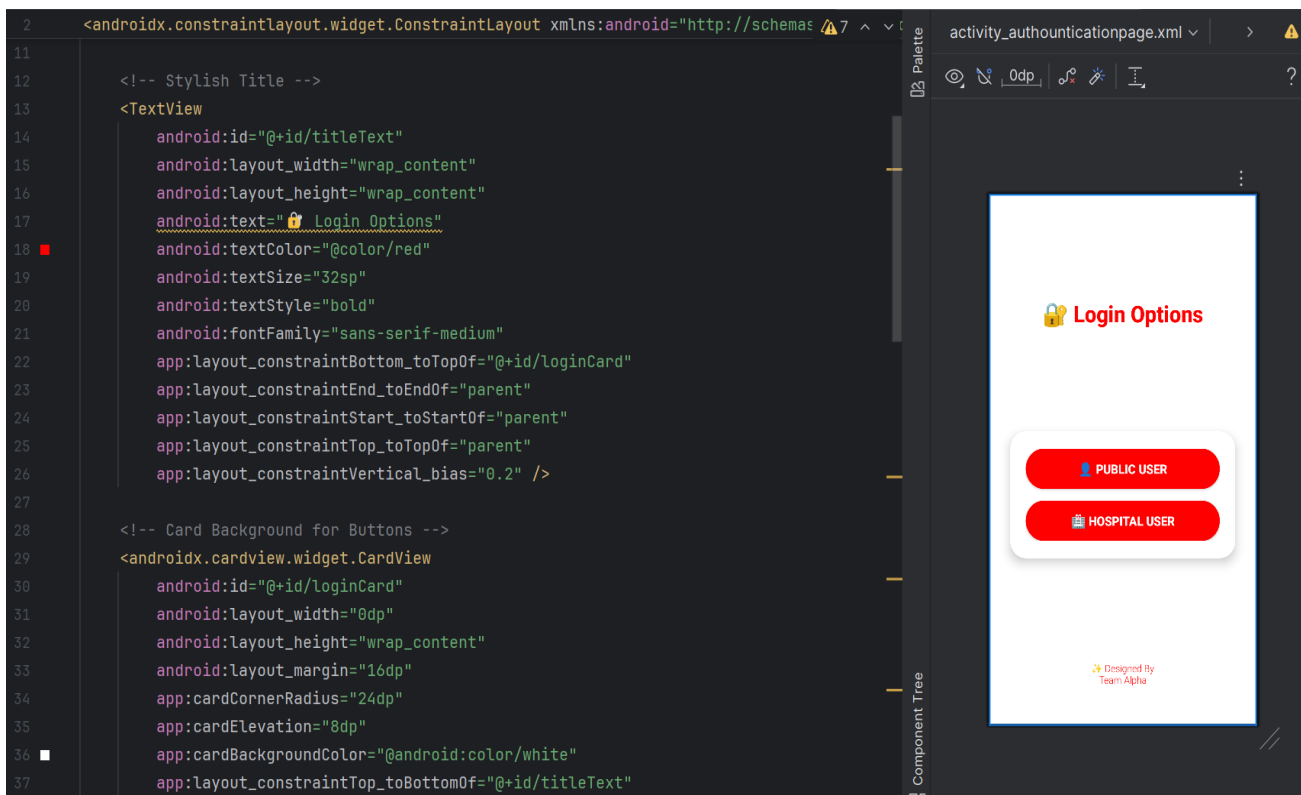


Fig. 5.2 – XML code of Login options page

5.2 Difficulties encountered and strategies used to tackle

Difficulties Encountered:

1. Firebase Setup and Configuration

Integrating multiple Firebase services (Authentication, Database, Maps) caused issues such as incorrect configuration files, outdated dependencies, and syncing errors.

2. Real-Time Data Handling

Displaying live donor information and updating UI components in real time was challenging, especially when multiple users were updating the database simultaneously.

3. Google Maps Integration

Generating a valid API key, enabling required APIs, and granting Android app permissions sometimes caused map loading errors or blank screens.

4. UI Alignment and Responsiveness

Ensuring that different screens (login, dashboard, request page) looked correct on various screen sizes required multiple adjustments in XML layouts.

5. Emulator & Device Testing Issues

The Android emulator was slow at times, sometimes causing crashes during testing, especially while testing Google Maps or location-based features.

6. Error Handling & Validation

Managing different cases like empty fields, incorrect data format, failed authentication, or network errors required extra validation logic.

Strategies Used to Overcome Difficulties:

1. Proper Firebase Configuration

- Re-installed the google-services.json file correctly.
- Synced Gradle frequently after adding dependencies.
- Used official Firebase documentation to resolve conflicts.

2. Using Listeners for Real-Time Data

- Implemented Firebase ValueEventListener and ChildEventListener to update donor lists instantly.
- Used Kotlin Coroutines where needed to avoid blocking the UI thread.

3. Fixing Google Maps Issues

- Enabled Maps, Geolocation, and Places API in Google Cloud Console.
- Tested API key restrictions to ensure the key worked only for the mobile app.
- Added necessary permissions like ACCESS_FINE_LOCATION in the Android Manifest.

4. Improving UI Compatibility

- Used ConstraintLayout for better arrangement of UI elements.
- Used wrap_content, match_parent, and layout guidelines for responsive design.
- Tested layouts on multiple emulator screen sizes.

5. Optimizing Emulator Testing

- Enabled hardware acceleration (HAXM).
- Used physical Android devices when emulator lag was too high.
- Cleared emulator cache and used lightweight virtual devices.

6. Adding Strong Validations

- Added field-level validation for email, password, blood group, and phone number.
- Implemented try-catch blocks for crash prevention.
- Displayed meaningful Toast messages to guide users.

5.3 Algorithms Used:

Some algorithms used were:

1. Authentication Algorithm (Firebase Authentication Flow)

Used for login and registration.

- User enters email/phone/password
- Input validation
- Firebase checks credentials
- If valid → login success
- If invalid → authentication error

2. User Input Validation Algorithm

Used to verify data before storing it:

- Check for empty fields
- Check valid email format
- Check password length
- Check valid phone number
- Check valid blood group
- If all conditions satisfied → proceed
- Else → show error

3. Filtering & Matching Algorithm (Blood Group Matching)

Used to find suitable donors:

- Compare user's required blood group with donor list
- Filter donors who match
- Display only compatible donors

CHAPTER 6

TESTING

6.1 Unit testing

Test Case ID	Module / Component	Test Description	Status
UT-01	Login Module	Check email format validation	Pass
UT-02	Registration	Test empty fields	Pass
UT-03	Firebase Auth	Verify user creation	Pass
UT-04	Database Module	Check data insertion	Pass
UT-05	UI Component	Test button clicks	Pass

Table 6.1

6.2 Integration testing

Test Case ID	Module / Component	Test Description	Status
IT-01	Login + Firebase Auth	Verify login integration	Pass
IT-02	Registration + Database	Save new user to database	Pass
IT-03	Maps + GPS	Load user location	Pass
IT-04	Donor List + Realtime DB	Fetch donors from Firebase	Pass
IT-05	Profile + Storage	Load stored profile image	Pass

Table 6.2

6.3 System testing

Test Case ID	Module / Component	Test Description	Status
ST-01	Complete Login Flow	Register → Login → Dashboard	Pass
ST-02	Emergency Blood Request	Create request	Pass
ST-03	Navigation & UI Testing	Navigate through all screens	Pass
ST-04	Cross-Device Testing	Run on different devices	Pass
ST-05	Internet Failure Handling	To check if app shows error on turning off internet	Pass
ST-06	Permissions	To check if app asks for permission when disabling GPS	Pass

Table 6.3

CHAPTER 7

RESULTS

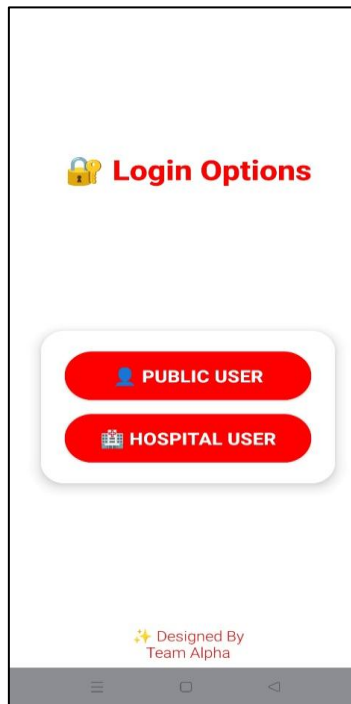


Fig 7.1 – Login Options

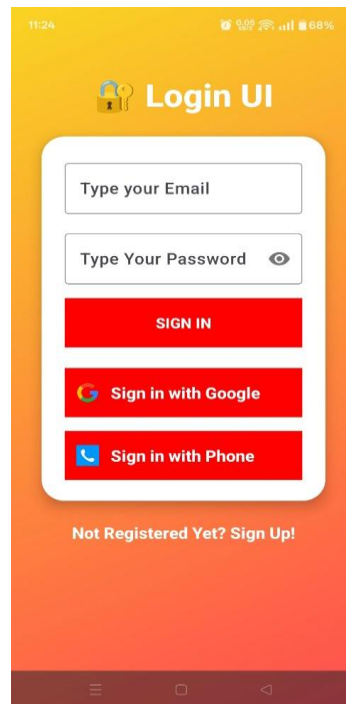


Fig 7.2 – Public User login

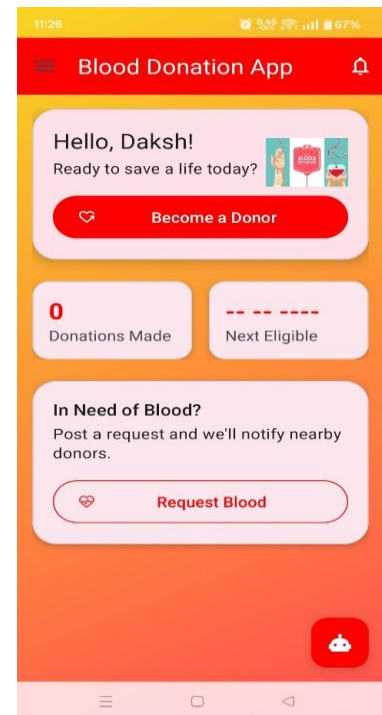


Fig 7.3 – Dashboard of Public user

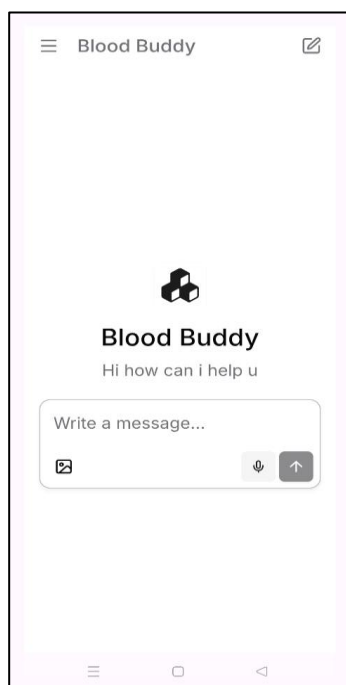


Fig 7.4 – AI chatbot support

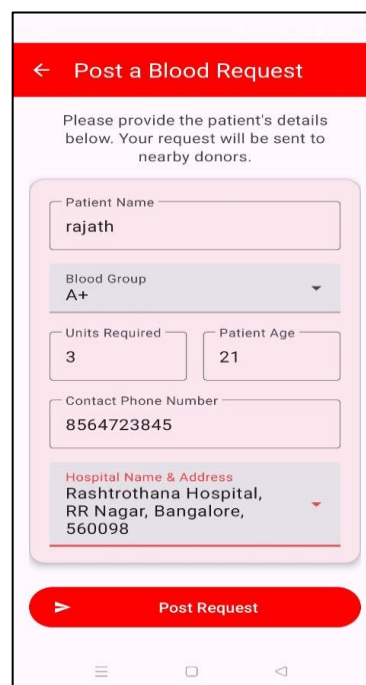


Fig 7.5 – Posting a blood Request

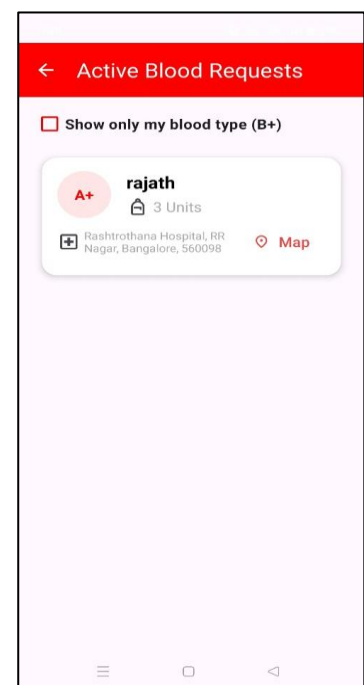


Fig 7.6 – Request appears on Donations Page

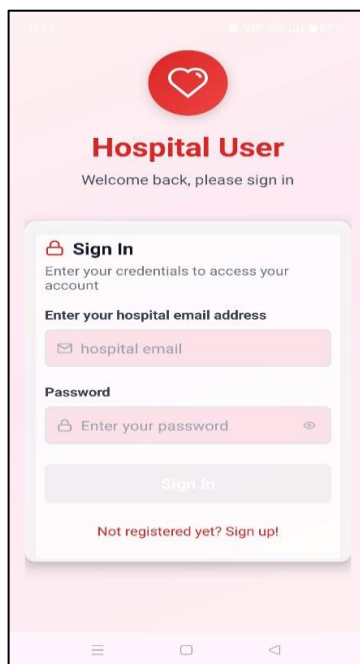


Fig 7.7 – Hospital User Login

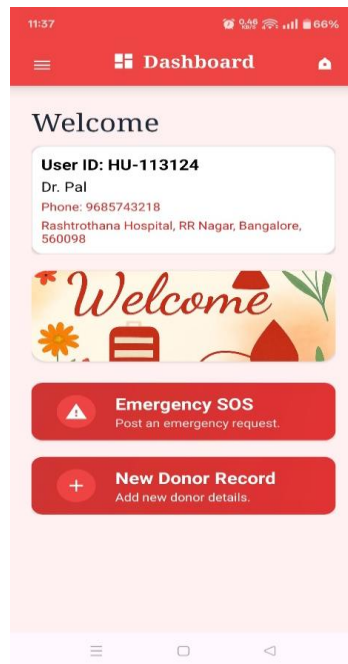


Fig 7.8 – Dashboard of Hospital User

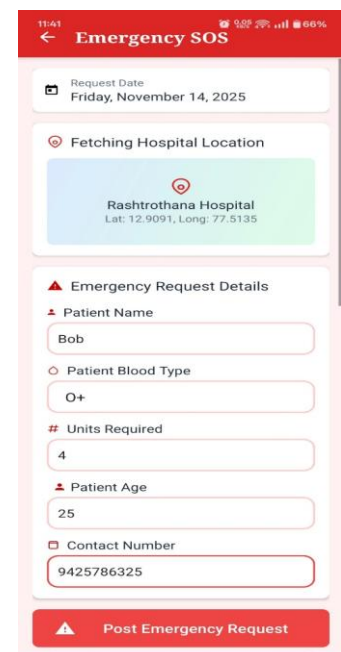


Fig 7.9 – Posting Emergency request

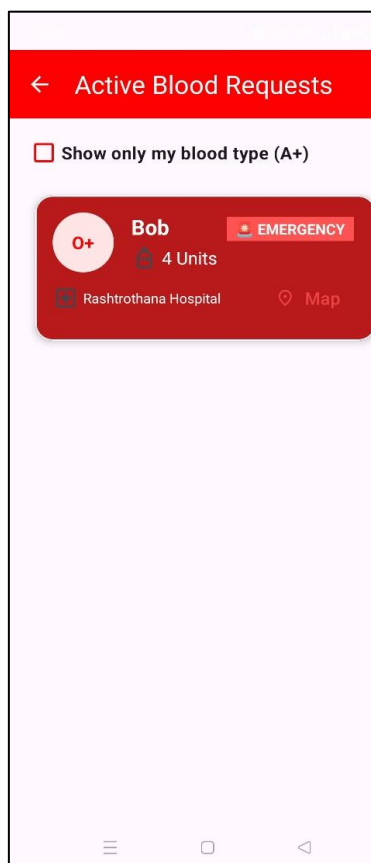


Fig 7.10 – Request appears on Donations Page

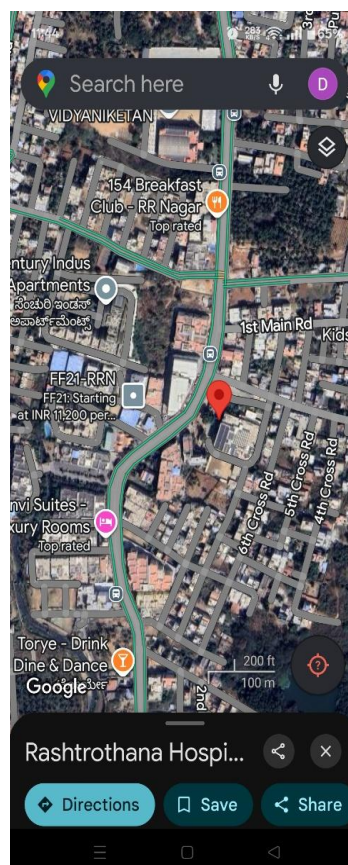


Fig 7.11 – Hospital Location on Google map



Fig 7.12 – Creating a record for donor

CHAPTER 8

CONCLUSION

8.1 Conclusion

The Blood Donation App – BloodBridge successfully addresses several key challenges found in traditional blood donation systems and existing mobile applications. By integrating essential features such as secure Firebase authentication, real-time database synchronization, and a clean and intuitive user interface, the application ensures reliability, accuracy, and ease of access for both donors and recipients.

The addition of enhanced features like an AI-powered chatbot, emergency SOS requests, and hospital-entered health records further improves user experience by providing quick assistance, faster communication, and accurate donor eligibility information. These components collectively make the system more efficient, user-friendly, and suitable for emergency healthcare scenarios.

Overall, the app provides a comprehensive platform that simplifies the donation process, minimizes delays during emergencies, and improves coordination between donors, hospitals, and individuals in need. With its robust architecture and stable performance, BloodBridge lays a strong foundation for future enhancements and large-scale deployment.

8.2 Future Scope of Work

While the current version of the application meets the core objectives, several advanced features can be incorporated to further enhance its functionality and real-world usability:

Real-Time Navigation & Live Tracking:

By integrating advanced Google Maps APIs, the app can allow hospitals or emergency coordinators to track the live location of donors who have accepted a blood request. This ensures faster contact, optimized travel routes, and timely arrival at the hospital.

Donor-Hospital Communication System:

A built-in chat or calling feature can help donors directly communicate with hospital staff for clarifications, directions, or health queries

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